Shiyu Liu

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Education

Renmin University of China *Sep. 2024 – Present*
Dual-Degree Program in Finance and Data Science & Big Data Technology
School of Information, Jinke Experimental Class Beijing, China

Research Interests

Machine Learning • Large Language Models & LLM Systems • Algorithmic Game Theory & Mechanism Design • Online Decision-Making & Optimization

Publications & Manuscripts

Peer-Reviewed Publications

[1] Yuqi Kong*, **Shiyu Liu***, Jiaxu Li, Hongtao Liu, Qi Qi, Weiran Shen. "Modeling and Solving Stable Matching under Probabilistic Preferences with Large Language Models." *ACL 2026*.
* Equal contribution.

Manuscripts Under Review

[2] **Shiyu Liu**, Weiran Shen. "SPARE: Selective Prefill Scheduling to Improve Throughput and Latency in LLM Serving." *Under Review*, 2026.

[3] Ren Liu, **Shiyu Liu**, Xiaodong Liu, Yuan Guo, Chen Yuanze, Lihua Zhang, Weiran Shen. "Revenue Optimization in Repeated Auctions with Additive Boosts." *Under Review*, 2026.

Research Experience

Renmin University of China | Advisor: Prof. Weiran Shen

SPARE: Selective Prefill Scheduling to Improve Throughput and Latency in LLM Serving

Undergraduate Researcher (Project Lead) Mar. 2026 – Present

- Investigated whether strategically delaying unfinished prefill computation could improve both first-token latency and latency-constrained serving capacity in memory-constrained LLM serving systems.
- **Led the design and implementation of SPARE**, a KV-cache-aware online scheduler that selectively defers unfinished prefills while protecting immediate output opportunities, using pressure-adaptive token budgets and chunk limits based on projected physical KV-block savings.
- Developed a **GPU-calibrated, vLLM V1-aligned evaluation framework**, achieving **99.24% feasibility agreement** with GPU serving across four LLMs; led experimental evaluation and manuscript preparation.
- Demonstrated **up to 2.65× serving capacity** under matched mean-TTFT constraints and **up to 82.1% lower mean TTFT** at matched arrival rates, showing that strategic deferral can mitigate KV-cache contention and improve the latency–capacity trade-off.

Revenue Optimization in Repeated Auctions with Additive Boosts

Undergraduate Researcher Feb. 2025 – Jun. 2025

- Investigated revenue optimization in repeated second-price auctions, where intertemporal incentives can undermine truthful bidding despite incentive compatibility in individual auction rounds.
- Contributed to the design of a **threat-based boost mechanism** that uses future boost allocations to align bidders' incentives, and participated in theoretical analysis and proofs of **Bayesian incentive compatibility**.
- **Led the implementation and experimental evaluation** of the proposed mechanism, including revenue optimization using analytical gradients and projected gradient ascent, and conducted experiments with synthetic and real-world auction data.
- Demonstrated revenue improvements of at least **19.30% and 21.76%** over the strongest static benchmarks for homogeneous and heterogeneous buyers, respectively; contributed to research discussions and manuscript preparation, highlighting how intertemporal incentives can improve revenue while preserving truthful bidding.

Research Experience (continued)

Modeling and Solving Stable Matching under Probabilistic Preferences with LLMs

Undergraduate Researcher (Co-First Author) Jun. 2025 – Jan. 2026

- Investigated stable matching under stochastic preferences, addressing the limitations of classical deterministic stability in capturing uncertain and boundedly rational decision-making.
- Designed an **LLM-based probabilistic preference framework** grounded in quantal response theory, introduced **Expected Blocking Pairs (EBP)** to quantify instability, and developed the **Hybrid GS–LLM matching algorithm** with probabilistic acceptance decisions.
- Led implementation, experimental design, and evaluation across **five LLMs and 21 real-world speed-dating groups**, analyzing LLM-generated preferences, switching behavior, and matching stability; authored the paper's Methodology section.
- Showed structured quantal-response patterns in LLM-induced decisions and **lower expected instability than classical Gale–Shapley in most evaluated groups**, demonstrating the value of uncertainty-aware matching mechanisms.

Research Grants

Beijing Municipal Natural Science Foundation – Undergraduate Qiyan Research Program 2026

Project Lead (Sole Student Investigator); *Dynamic Scheduling and Batching Optimization for LLM Serving.*

National Undergraduate Innovation Training Program

Project Lead; national-level approved research training project.

National Undergraduate Entrepreneurship Training Program

Project Lead; national-level approved entrepreneurship training project.

Honors & Awards

2026 **CCF Undergraduate Qihang Program** – Selected as one of 101 students nationwide from 56 universities.

2026 **First Prize (Final Round), 17th JD Cup Student Innovation and Entrepreneurship Competition** – CheeseX, an AI-driven university project collaboration platform; one of four first-prize teams among 74 entries.

2026 **Third-Class Academic Scholarship**, Renmin University of China.

2025 **Three-Good Student Award**, Renmin University of China.

2024 **Meritorious Winner, Mathematical Contest in Modeling (MCM)** – Awarded to approximately 6.3% of 2024 MCM participating teams.

Third Prize, Chinese Mathematics Competition for College Students (Beijing Division) – Municipal-level award.

Selected Talks & Presentations

Student Representative & Speaker, CCF Qihang Program 2026

Delivered a speech on behalf of the undergraduate cohort at a CCF event, sharing research experiences and reflections on academic development.

Speaker, AI + Learning Academic Seminar Series 2026

Delivered student-oriented seminars on artificial intelligence, large language models, and practical AI tools, combining foundational concepts with hands-on demonstrations.

Technical Skills

AI-Native Research & Development: Advanced proficiency in OpenAI Codex CLI, Claude Code, and Cursor; AI-assisted codebase exploration, software development, debugging, and research experimentation.

Agentic Systems & Workflows: Model Context Protocol (MCP), tool/function calling, LLM API integration, agent orchestration, and multi-step AI workflows.

Machine Learning & LLM Systems: PyTorch, Hugging Face Transformers, vLLM; GPU-based benchmarking and experimental evaluation.

Programming & Development: Python, C/C++, SQL; Linux, Git/GitHub, Docker, OpenMP.

Research Tools: LaTeX, VS Code.